

# **Lecture 5: Atmospheres I**

## **Composition, Structure, & Energy Balance**

### **Planetary Systems**

A horizontal, slightly curved blue line with a soft glow, representing the Earth's atmosphere as seen from space. It is centered behind the text.

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2026

## Learning objectives

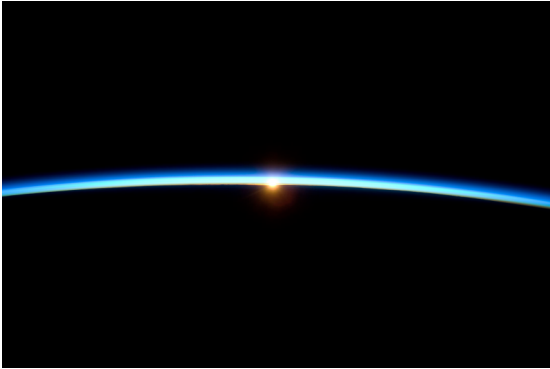
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By the end of this lecture, you will be able to:

- Classify atmospheric types (primary, secondary, tertiary)
- Derive pressure profiles from hydrostatic equilibrium
- Compute the atmospheric scale height and apply it across the solar system
- Explain how the greenhouse effect raises surface temperature above  $T_{\text{eff}}$
- Evaluate atmospheric escape mechanisms and retention criteria

## Recap: Lecture 4

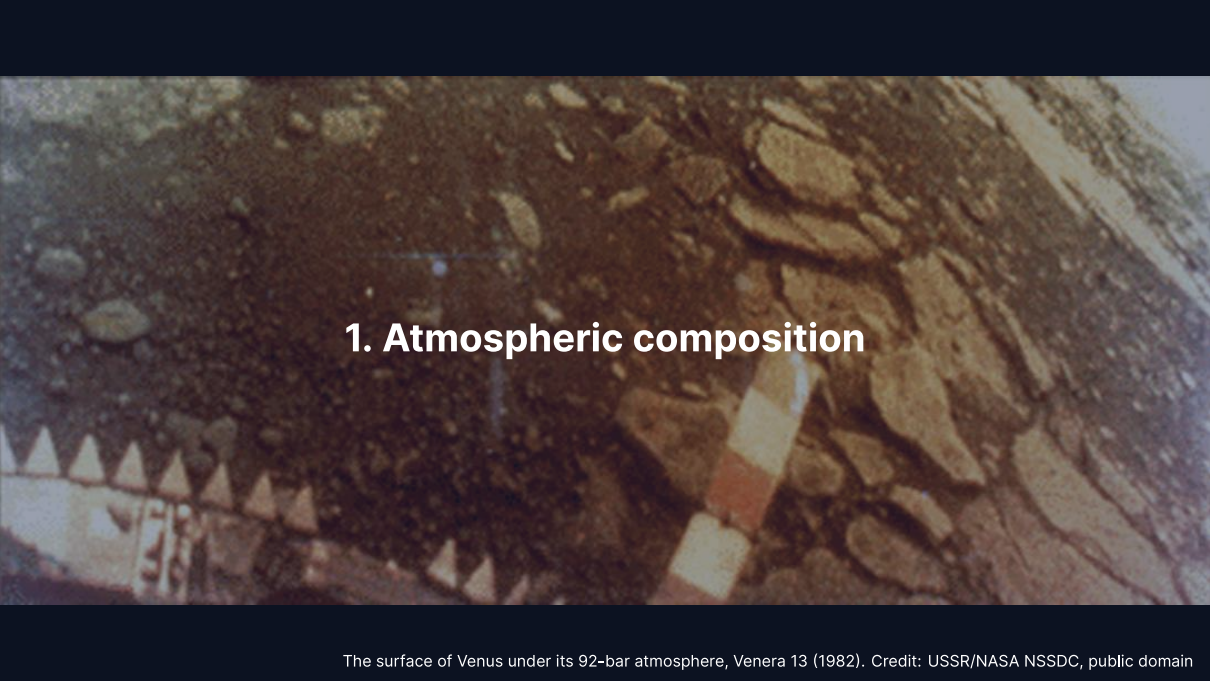
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Credit: NASA/ISS Expedition 21 crew, public domain

- Metal-silicate separation in magma oceans drove core formation within ~30–60 Myr
- Dynamo requires conducting fluid, convection, and magnetic Reynolds number  $R_m \gtrsim 10\text{--}100$
- Mars lost its dynamo between ~4.1 and 3.7 Ga, triggering atmospheric erosion

**Today:** The thin gaseous envelope that controls a planet's surface conditions.



## 1. Atmospheric composition

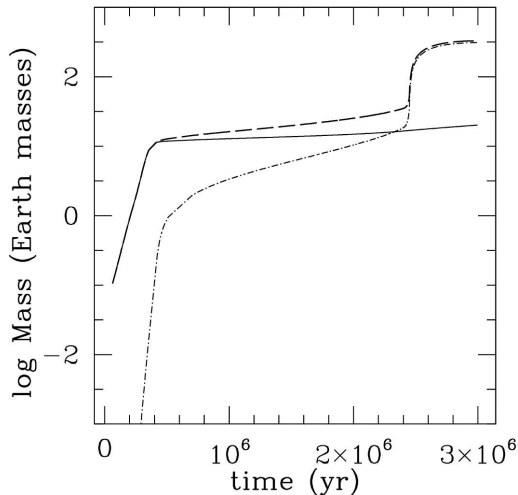
The surface of Venus under its 92-bar atmosphere, Venera 13 (1982). Credit: USSR/NASA NSSDC, public domain

# Primary atmospheres

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## Captured directly from the protoplanetary disk.

- Dominated by  $\text{H}_2$  and He (solar composition)
- Requires core mass  $\gtrsim 5\text{--}10 M_{\oplus}$  before disk dispersal in  $\sim 3\text{--}10$  Myr
- **Gas giants:** Jupiter and Saturn retain massive  $\text{H}_2/\text{He}$  envelopes
- **Ice giants:** Uranus and Neptune envelopes are only 10–20% of total mass
- Terrestrial planets are too small to retain primordial  $\text{H}_2/\text{He}$



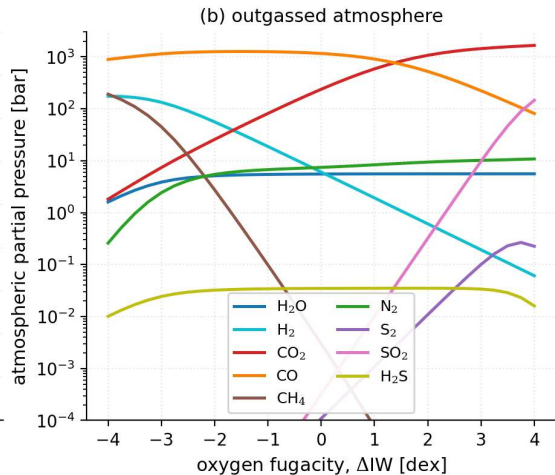
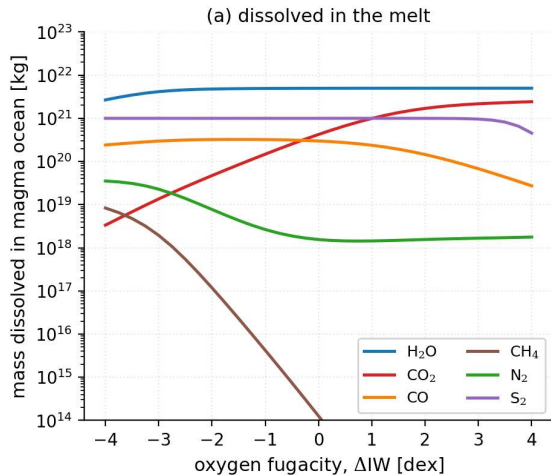
Growth of a giant planet by core accretion: runaway gas accretion builds the  $\text{H}_2/\text{He}$  envelope within the disk lifetime once envelope mass overtakes core mass. Reproduced from Helled et al. (2014), Fig. 1; simulation from Lissauer et al. (2009).

## Secondary atmospheres

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### Produced by outgassing from the interior.

- Magma ocean degassing and volcanism release dissolved volatiles
- Speciation depends on mantle **oxygen fugacity** ( $f_{\text{O}_2}$  redox state):
  - Oxidising:  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ ,  $\text{N}_2$
  - Reducing:  $\text{H}_2$ ,  $\text{CO}$ ,  $\text{N}_2$
- **Venus:**  $\text{CO}_2$ -dominated (96.5%), 92 bar
- **Mars:**  $\text{CO}_2$ -dominated (95%), 6 mbar
- **Titan:** thick  $\text{N}_2$  atmosphere (from  $\text{NH}_3$ ), 1.5 bar



Speciation of an outgassed atmosphere as a function of melt oxygen fugacity:  $CO$ ,  $H_2$ , and  $CH_4$  dominate under reducing conditions, while  $CO_2$  and  $H_2O$  dominate under oxidising conditions. Course figure; model of Nicholls et al. (2024).



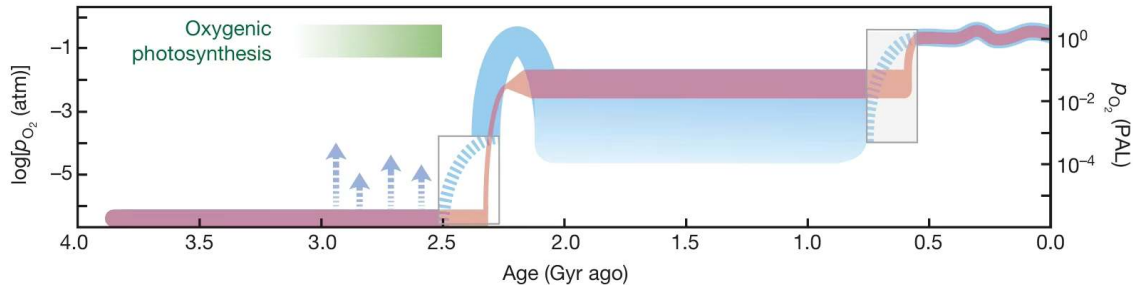
## Tertiary atmospheres

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Earth's atmosphere is a tertiary atmosphere produced by billions of years of biological modification.

- Original outgassed inventory was dominated by  $\text{CO}_2$  and  $\text{N}_2$ , similar to Venus
- Oxygenic photosynthesis added the  $\text{O}_2$  (21%); the  $\text{N}_2$  (78%) is outgassed and accumulated
- Atmospheric  $\text{O}_2$  is entirely biogenic and requires active life to persist

Earth's  $\text{O}_2$ , and so its tertiary character, is continuously maintained by life.



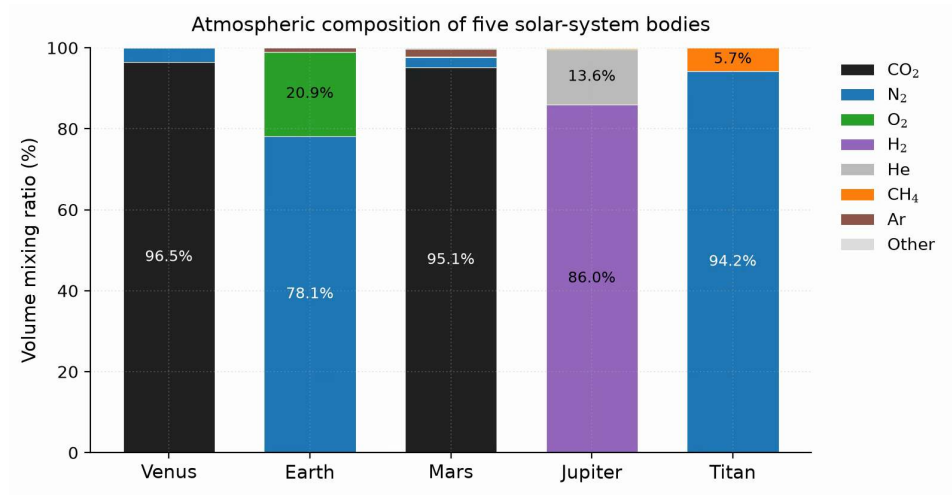
Earth's atmospheric oxygen through time: anoxic for its first ~2 Gyr, rising abruptly at the Great Oxidation Event near 2.4 Ga, and reaching modern levels in the Phanerozoic through biological photosynthesis. Reproduced from Lyons et al. (2014).

## Comparative atmospheric properties

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|                         | Venus           | Earth          | Mars            | Jupiter        | Titan           |
|-------------------------|-----------------|----------------|-----------------|----------------|-----------------|
| $P_{\text{surf}}$ (bar) | 92              | 1.0            | 0.006           | n/a            | 1.5             |
| $T_{\text{surf}}$ (K)   | 737             | 288            | 215             | n/a            | 94              |
| Dominant gas            | CO <sub>2</sub> | N <sub>2</sub> | CO <sub>2</sub> | H <sub>2</sub> | N <sub>2</sub>  |
| Secondary gas           | N <sub>2</sub>  | O <sub>2</sub> | N <sub>2</sub>  | He             | CH <sub>4</sub> |
| $\mu$                   | 43.4            | 28.97          | 43.3            | 2.2            | 28.6            |
| Type                    | 2 <sup>+</sup>  | 3 <sup>+</sup> | 2 <sup>+</sup>  | 1 <sup>+</sup> | 2 <sup>+</sup>  |

Staggering diversity: surface pressures spanning five orders of magnitude, temperatures from 94 to 737 K.

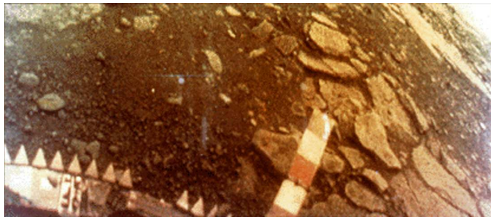


Atmospheric composition by volume for five solar-system bodies: three origins yield five distinct outcomes across the solar system. Course figure.

## Venus: a secondary atmosphere in extremis

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A massive secondary atmosphere transforms planetary surface conditions to extreme pressures and temperatures.



Credit: USSR/NASA NSSDC, public domain

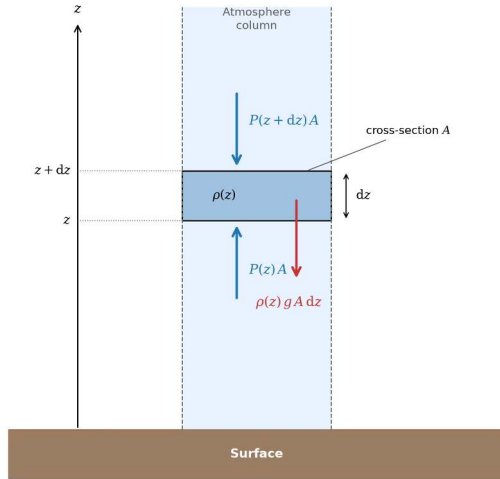
- *Venera 13* landed 1 March 1982 and survived 127 minutes
- Surface pressure of 92 bar and surface temperature of 737 K
- Dense CO<sub>2</sub> atmosphere and cloud deck filter sunlight to a dim orange glow



## 2. Hydrostatic equilibrium

The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL

# Hydrostatic equilibrium



Credit: Course figure

Net vertical force on a thin slab of area  $A$  and thickness  $dz$  vanishes:

$$P(z)A - P(z + dz)A = \rho g A dz$$

$$\frac{dP}{dz} = -\rho g$$

Pressure decreases upward because each layer supports the weight of all gas above it.

## The ideal gas law and barometric formula

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The ideal gas law relates density to pressure:  $P = \rho k_B T / (\mu m_u)$ .

Substituting  $\rho(P)$  into hydrostatic equilibrium  $\frac{dP}{dz} = -\rho g$ :

$$\frac{dP}{dz} = -\frac{\mu m_u g}{k_B T} P = -\frac{P}{H}$$

For an **isothermal** atmosphere (constant  $T$ ), integrating yields:

$$P(z) = P_0 \exp\left(-\frac{z}{H}\right)$$

where  $H = k_B T / (\mu m_u g)$ ; pressure drops by  $e \approx 2.718$  each scale height.





### **3. Blackboard derivation: scale height**

## Deriving the scale height

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**Goal:** Derive  $H = k_B T / (\mu m_u g)$  from hydrostatic equilibrium and the ideal gas law.

Start from:

$$\frac{dP}{dz} = -\rho g \quad \text{and} \quad P = \frac{\rho k_B T}{\mu m_u}$$

Express  $\rho$  in terms of  $P$ :

$$\rho = \frac{P \mu m_u}{k_B T}$$

Substitute:

$$\frac{dP}{dz} = -\frac{\mu m_u g}{k_B T} P \equiv -\frac{P}{H}$$

# The atmospheric scale height

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$$H = \frac{k_B T}{\mu m_u g}$$

Physical interpretation:

- **Higher**  $T \rightarrow$  larger  $H$ : hotter gas extends further
- **Heavier molecules** (larger  $\mu$ )  $\rightarrow$  smaller  $H$ : harder to loft
- **Stronger gravity**  $g \rightarrow$  smaller  $H$ : more compressed

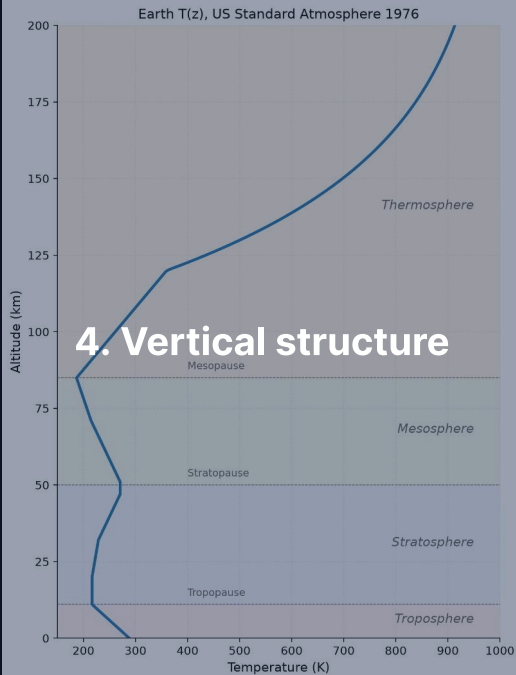
**Key insight:**  $H$  encodes the competition between thermal energy and gravitational binding; integrated over the whole column, the surface pressure is the column weight,  $P_0 = m_{\text{col}} g$ .

## Scale heights across the solar system

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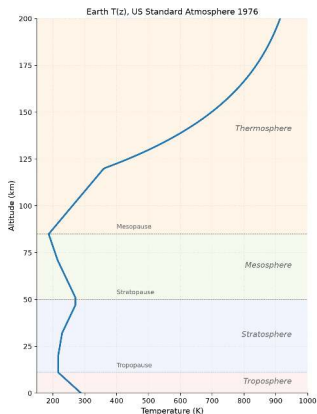
| Body    | $T$ (K) | $\mu$ | $g$ ( $\text{m s}^{-2}$ ) | $H$ (km) |
|---------|---------|-------|---------------------------|----------|
| Venus   | 737     | 43.4  | 8.87                      | 15.9     |
| Earth   | 288     | 28.97 | 9.81                      | 8.4      |
| Mars    | 215     | 43.3  | 3.72                      | 11.1     |
| Jupiter | 165     | 2.2   | 24.8                      | 25       |
| Titan   | 94      | 28.6  | 1.35                      | 20       |

- Jupiter: large  $H$  despite strong gravity, because  $\text{H}_2$  is very light ( $\mu = 2.2$ )
- Titan: large  $H$  despite low  $T$ , thanks to very weak gravity ( $g = 1.35 \text{ m s}^{-2}$ )
- Venus: hot but heavy molecules and decent gravity yield moderate  $H$



Earth's atmospheric layers. Course figure

# Atmospheric layers



Credit: Course figure; US Standard Atmosphere 1976

Layers defined by the sign of  $\frac{dT}{dz}$ :

1. **Troposphere** (0–12 km):  $T$  decreases with  $z$ ; convective
2. **Stratosphere** (12–50 km):  $T$  increases; UV ozone heating
3. **Mesosphere** (50–85 km):  $T$  decreases to mesopause (~190 K)
4. **Thermosphere** (>85 km):  $T$  increases; EUV heating ( $T > 1000$  K)

## The adiabatic lapse rate

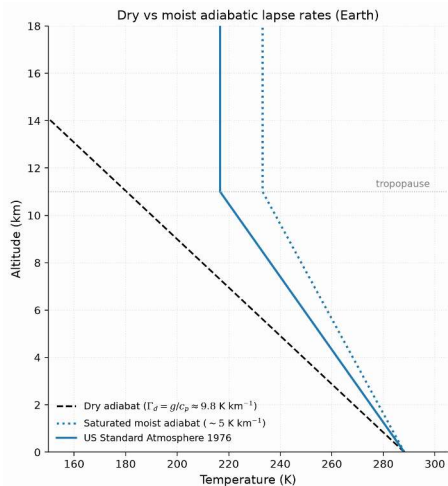
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In the troposphere, convection sets the temperature gradient:

$$\Gamma_d = -\frac{dT}{dz} = \frac{g}{c_p}$$

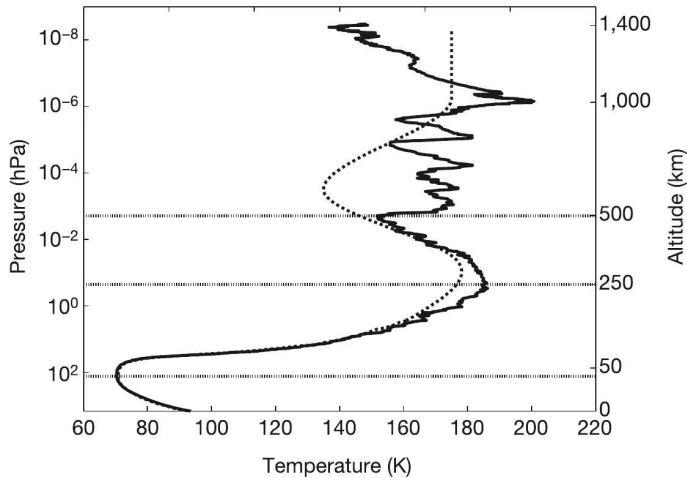
- **Dry adiabatic lapse rate** for Earth:  $\Gamma_d = 9.81/1004 \approx 9.8 \text{ K km}^{-1}$
- Observed average:  $\sim 6.5 \text{ K km}^{-1}$  (lower due to latent heat release)
- Physical picture: rising air expands, does work on surroundings, cools

**Tropopause height:**  $\sim 8 \text{ km}$  (poles) to  $\sim 17 \text{ km}$  (equator).



Dry and moist adiabats against the US Standard Atmosphere 1976 reference profile: latent heat makes the moist adiabat shallower, and real convection sits between both limits. Course figure.

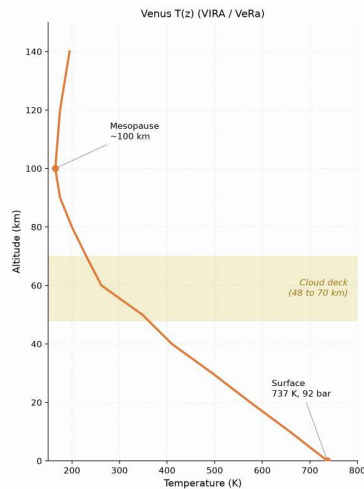




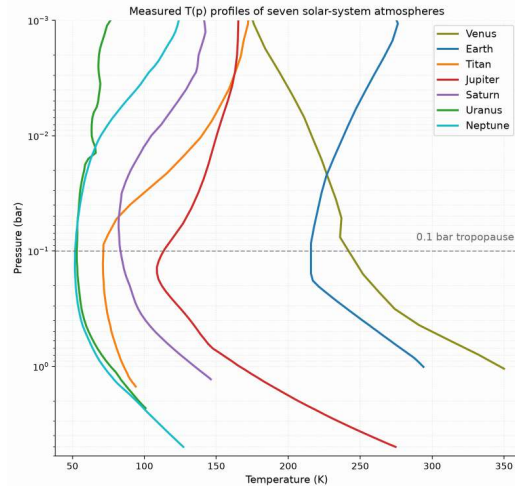
Titan's temperature profile from the Huygens HASI descent: an in situ descent measured the 94 K surface, 0.1 bar tropopause, and stratospheric structure. Fulchignoni et al. (2005).



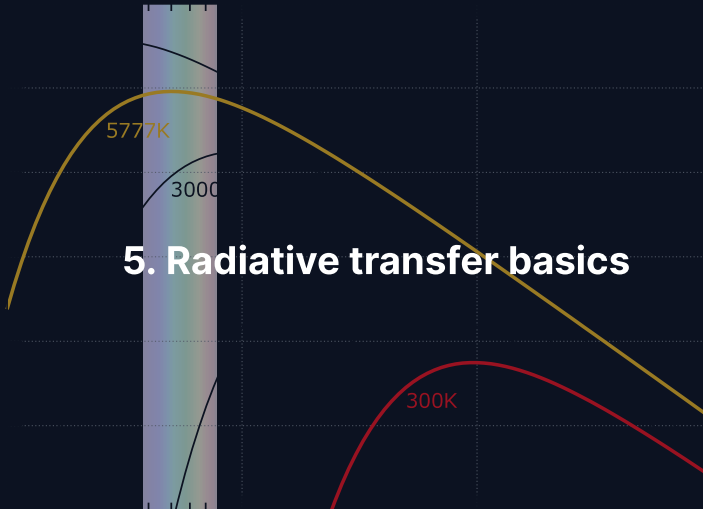
Detached haze layers in Titan's upper atmosphere imaged at the limb by Cassini ISS: photochemical haze layers extending up to 500 km altitude absorb solar UV and heat the stratosphere. Credit: NASA/JPL/Space Science Institute, public domain.



Venus temperature profile, Pioneer Venus/VIRA and Venus Express: without ozone, temperature falls monotonically from 737 K at the surface to the mesopause, reaching 1 bar near 50 km. Course figure.



Measured temperature-pressure profiles of seven solar-system atmospheres: thick atmospheres transition from convective to radiative near 0.1 bar, setting a universal tropopause minimum. Digitized from Robinson & Catling (2014), Fig. 1.



## Three interactions: radiation and matter

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When radiation passes through an atmosphere:

**Absorption:** Photon energy converted to internal molecular energy

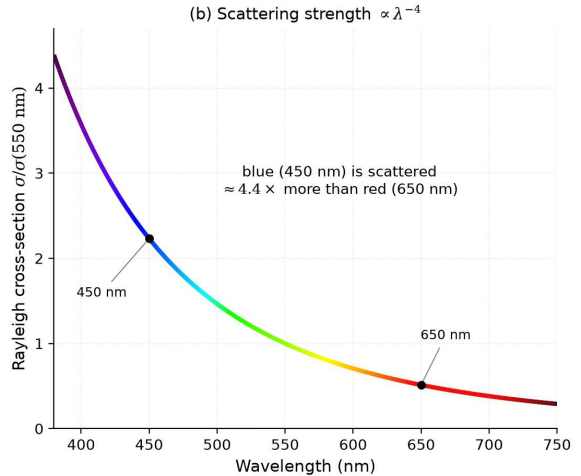
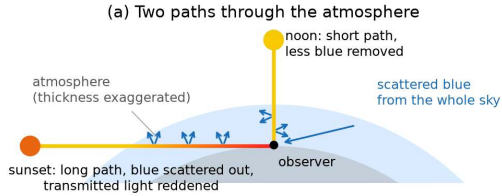
Key absorbers:  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ ,  $\text{CH}_4$ ,  $\text{O}_3$ ,  $\text{N}_2\text{O}$

**Emission:** Warm gas radiates at absorbed wavelengths (Kirchhoff's law)

Radiates energy directly to space

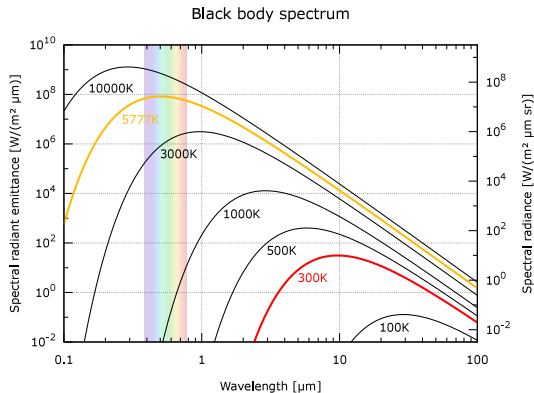
**Scattering:** Photons redirected without absorption

Rayleigh scattering ( $\propto \lambda^{-4}$ ) and Mie scattering



Rayleigh scattering and the colour of the sky: shorter path lengths at noon preserve blue scattered light, while longer slant paths at sunset transmit only reddened light; cross-section scales as  $\lambda^{-4}$ . Course figure.

# Stellar vs. planetary emission



Credit: Wikimedia, public domain

- Sun ( $\sim 5800 \text{ K}$ ): peak at  $\sim 0.5 \mu\text{m}$  (visible)
- Planet ( $\sim 300 \text{ K}$ ): peak at  $\sim 10 \mu\text{m}$  (thermal IR)

This **wavelength separation** is the physical basis of the greenhouse effect: the atmosphere can be transparent at visible wavelengths but opaque in the infrared.



## Optical depth and the Beer-Lambert law

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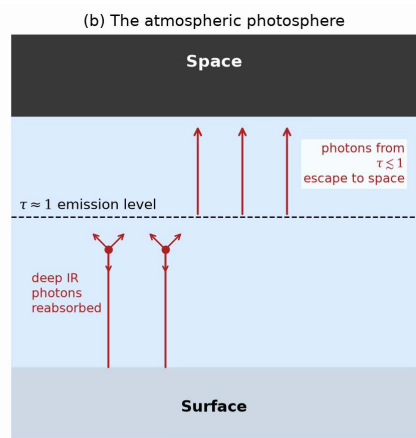
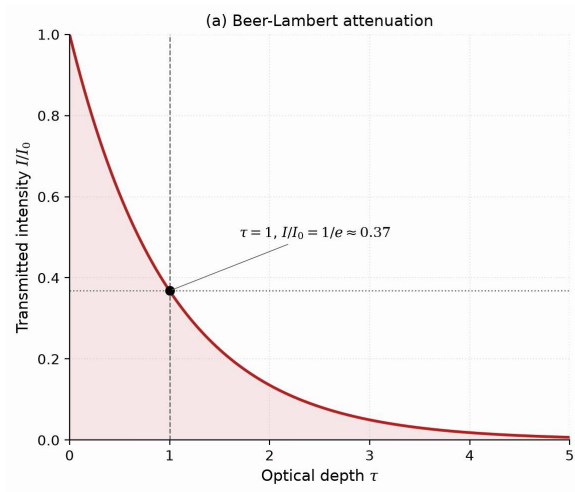
Optical depth  $\tau$  measures cumulative absorption along a path:

$$\tau = \int_0^s \kappa \rho \, ds'$$

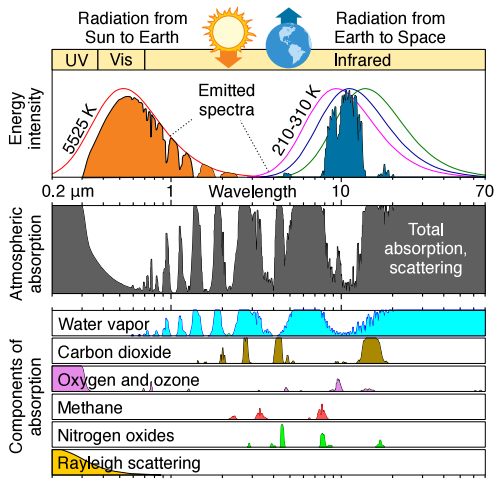
Beam intensity decays following the **Beer-Lambert law**:

$$I = I_0 e^{-\tau}$$

- **Optically thin** ( $\tau \ll 1$ ): radiation passes freely through the gas
- **Optically thick** ( $\tau \gg 1$ ): photons are repeatedly absorbed and re-emitted
- **Photosphere** ( $\tau \approx 1$ ): thermal radiation escapes directly to space

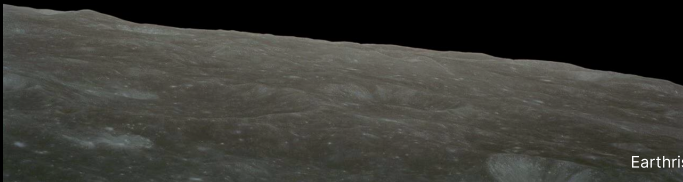


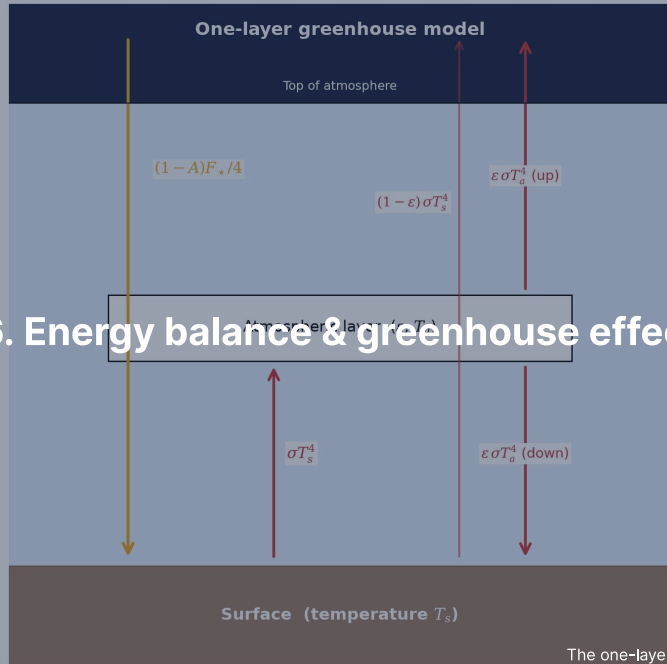
The atmospheric photosphere: thermal radiation escapes to space from the  $\tau \approx 1$  level, setting the effective temperature. Course figure.



Earth's atmospheric absorption spectrum from UV to IR: greenhouse gases absorb outgoing surface infrared while leaving visible sunlight and the 8–13  $\mu\text{m}$  window transparent. Credit: Wikimedia Commons, public domain.

**Break**





## 6. Energy balance & greenhouse effect

# Radiative-convective equilibrium

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Atmospheres divide into a convective lower layer and a radiatively controlled upper layer.

- **Troposphere:** greenhouse opacity drives convective instability along the adiabat
- **Stratosphere:** optically thin gas reaches radiative equilibrium, stable against convection
- **Tropopause:** boundary where the radiative temperature gradient becomes shallower than the adiabat

## Planetary energy balance

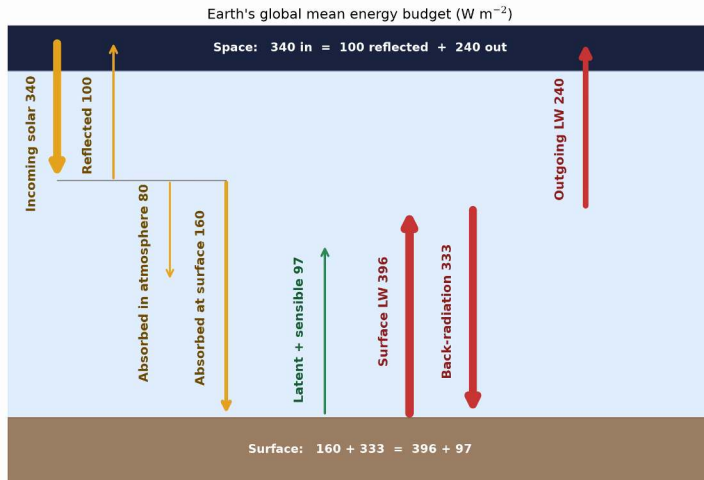
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Absorbed stellar power equals emitted thermal radiation:

$$(1 - A) \frac{S}{4} = \sigma T_{\text{eff}}^4 \quad \text{where} \quad S = \frac{L_{\star}}{4\pi d^2}$$

- $A$ : Bond albedo;  $S$ : stellar flux ( $S_0 = 1361 \text{ W m}^{-2}$  at Earth)
- Factor  $1/4$ : ratio of intercepting cross-section  $\pi R_p^2$  to emitting sphere  $4\pi R_p^2$

$$T_{\text{eff}} = \left[ \frac{(1 - A) L_{\star}}{16\pi\sigma d^2} \right]^{1/4} = \left[ \frac{(1 - A) S}{4\sigma} \right]^{1/4}$$



Earth's globally averaged energy budget in  $\text{W m}^{-2}$ : atmospheric back-radiation of  $333 \text{ W m}^{-2}$  supplies more energy to the surface than direct sunlight ( $160 \text{ W m}^{-2}$ ). After Trenberth et al. (2009).



## Effective vs. surface temperature

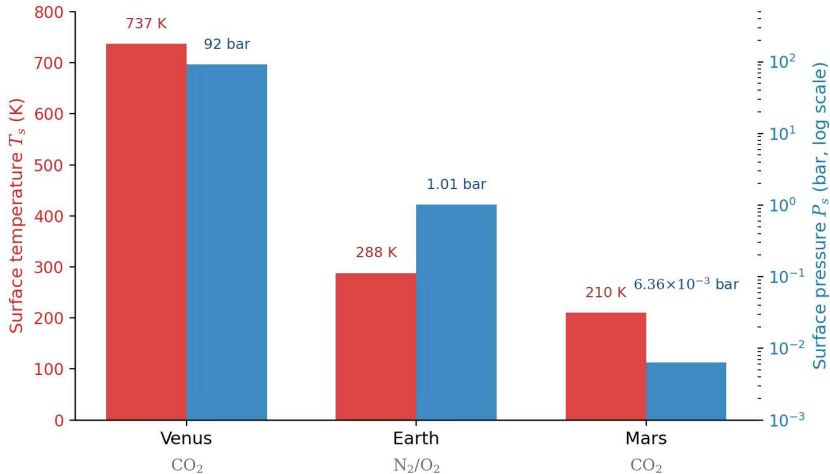
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| Body    | A    | d (AU) | $T_{\text{eff}}$ (K) | $T_{\text{surf}}$ (K) | $\Delta T$ (K) |
|---------|------|--------|----------------------|-----------------------|----------------|
| Venus   | 0.77 | 0.72   | 227                  | 737                   | +510           |
| Earth   | 0.30 | 1.00   | 255                  | 288                   | +33            |
| Mars    | 0.25 | 1.52   | 210                  | 215                   | +5             |
| Jupiter | 0.34 | 5.20   | 110                  | 165*                  | +55            |

\*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

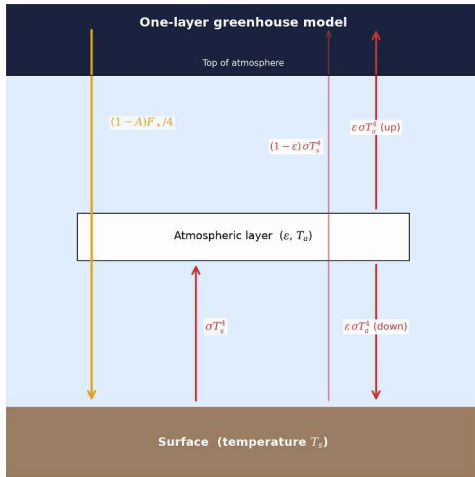
$\Delta T = T_{\text{surf}} - T_{\text{eff}}$  reveals the **greenhouse effect**.

Venus: 510 K. Earth: 33 K. Mars: ~5 K (only 6 mbar, no water-vapour amplifier).



Surface temperature and surface pressure of Venus, Earth, and Mars: planets of similar bulk composition span a factor of  $10^4$  in surface pressure and 500 K in surface temperature due to greenhouse warming. Course figure; data from NASA.

# The greenhouse mechanism



Credit: Course figure

1. Sunlight (visible,  $\sim 0.5 \mu\text{m}$ ) reaches the surface
2. Surface emits thermal IR ( $\sim 10\text{--}15 \mu\text{m}$ )
3. Greenhouse gases absorb outgoing IR
4. Absorbing layer re-emits: half up, half down
5. Downward emission adds energy to surface

**Result:**  $T_{\text{surf}} > T_{\text{eff}}$

# One-layer greenhouse model

Single atmospheric layer with infrared emissivity  $\varepsilon$ , transparent to visible sunlight:

## Flux balance:

Atmosphere:  $\varepsilon\sigma T_s^4 = 2\varepsilon\sigma T_a^4$

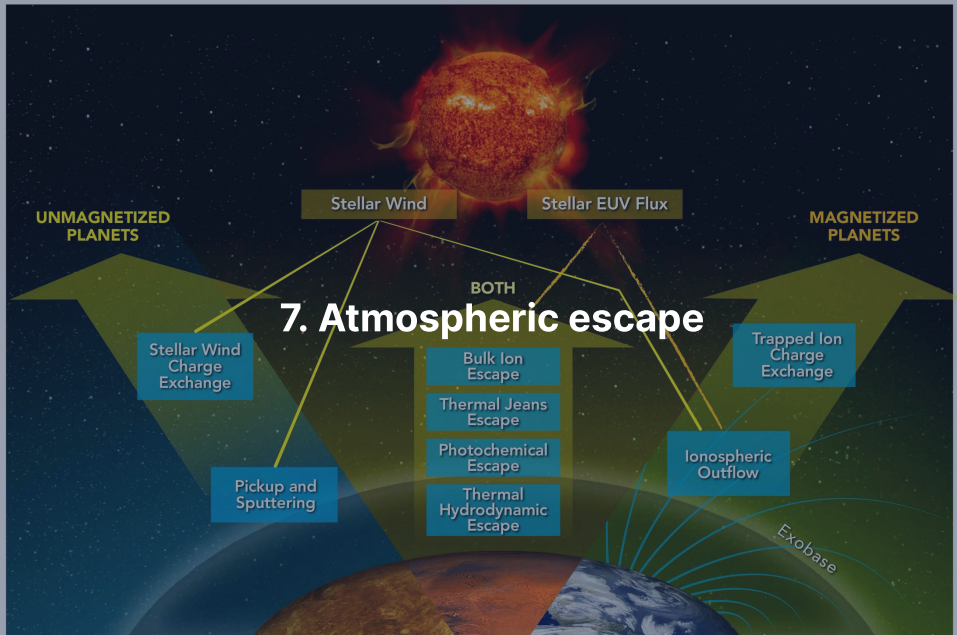
Surface:  $(1 - A)\frac{F_\star}{4} + \varepsilon\sigma T_a^4 = \sigma T_s^4$

TOA:  $(1 - A)\frac{F_\star}{4} = \sigma T_{\text{eff}}^4$

- $\varepsilon = 0$  (transparent):  $T_s = T_{\text{eff}}$
- $\varepsilon = 1$  (opaque):  $T_s = 2^{1/4} T_{\text{eff}} \approx 1.19 T_{\text{eff}}$  (Earth:  $\approx 303$  K)

## Surface temperature:

$$T_s = T_{\text{eff}} \left( \frac{2}{2 - \varepsilon} \right)^{1/4}$$



## Thermal (Jeans) escape

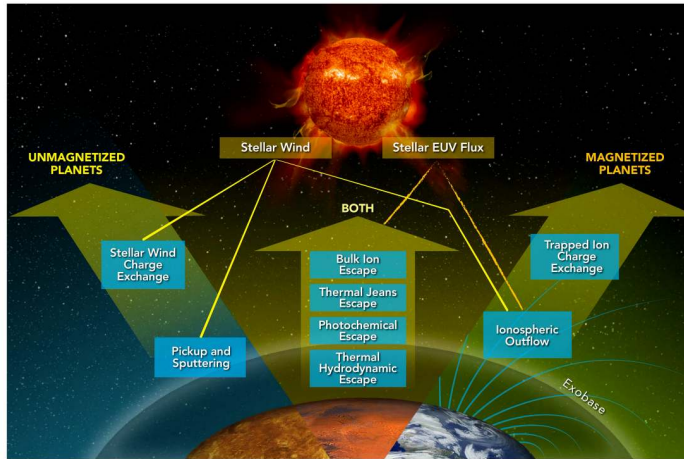
Molecules follow a Maxwell-Boltzmann distribution whose high-velocity tail can exceed escape speed.

The **Jeans escape parameter** at the exobase compares gravitational to thermal energy:

$$\lambda_J = \frac{GM_p m}{k_B T_{\text{exo}} r_{\text{exo}}} = \frac{v_{\text{esc}}^2}{v_{\text{th}}^2}$$

with  $v_{\text{th}} = \sqrt{2k_B T_{\text{exo}}/m}$  the most probable thermal speed.

- $\lambda_J \gg 1$ : escape is negligible because the tail is depleted
- $\lambda_J \lesssim 2-3$ : thermal escape drives rapid atmospheric loss
- Escape flux  $\Phi_J \propto (1 + \lambda_J) e^{-\lambda_J}$  depends exponentially on  $\lambda_J$



The main atmospheric escape processes: thermal, photochemical, and ion channels operate on all planets, while planetary magnetic fields select which pathways dominate. Gronoff et al. (2020).

## Jeans parameter for Earth and Mars

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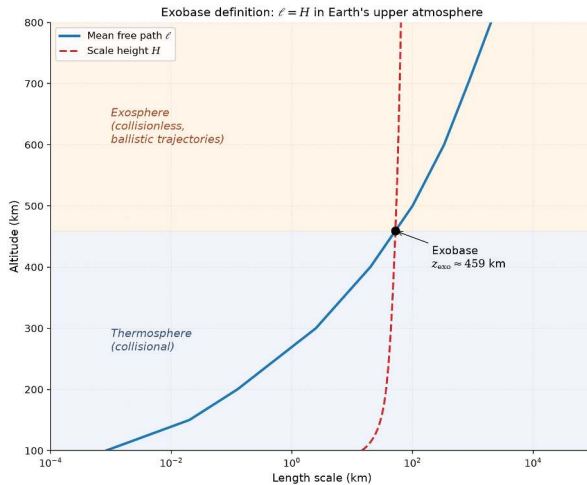
Because  $\lambda_j \propto m$ , light hydrogen escapes steadily while heavy molecules remain gravitationally bound.

Exobase temperatures: 1000 K (Earth), 270 K (Mars).

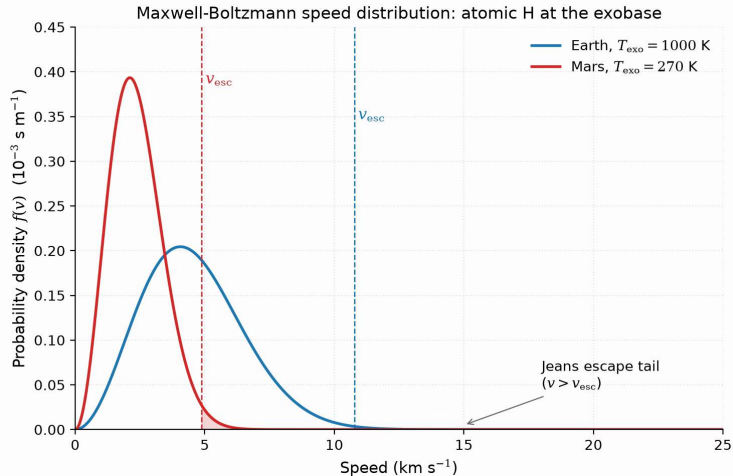
| Species         | $m$ (u) | $\lambda_j$ (Earth) | $\lambda_j$ (Mars) |
|-----------------|---------|---------------------|--------------------|
| H               | 1       | 7.0                 | 5.3                |
| H <sub>2</sub>  | 2       | 14                  | 11                 |
| He              | 4       | 28                  | 21                 |
| N <sub>2</sub>  | 28      | 200                 | 150                |
| CO <sub>2</sub> | 44      | 310                 | 230                |

- $\lambda_j \gg 1$  for N<sub>2</sub> and CO<sub>2</sub>: thermal escape is completely negligible
- Moderate  $\lambda_j$  allows atomic H to escape steadily from both planets





The exobase: altitude where the mean free path  $\ell$  equals the scale height  $H$ , splitting collisional fluid below from ballistic flight above. Course figure.



Maxwell-Boltzmann speed distributions for atomic H at Earth and Mars exobases: the escaping fraction sits in the high-velocity tail exceeding  $v_{\text{esc}}$ , scaling as  $e^{-\lambda}$ . Course figure.

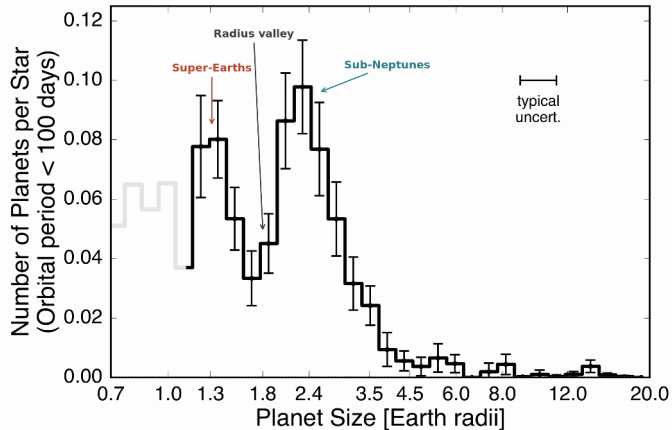
## Hydrodynamic escape

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Intense stellar EUV heating drives bulk outflow as an **atmospheric wind**:

- Peak importance in first few hundred Myr when stellar EUV is 10–100× higher
- Strips  $\text{H}_2$ -rich primary envelopes from planets up to several  $M_{\oplus}$
- Outflowing light hydrogen can drag along heavier volatiles (He, C, N, O)
- Primary mechanism sculpting the exoplanet radius valley at  $\sim 1.5\text{--}2 R_{\oplus}$

Source: Hunten et al. (1987); Owen (2019)



The observed radius valley in the California-Kepler Survey: a bimodal distribution with a deficit near  $1.8 R_{\oplus}$  separates stripped rocky cores from envelope-retaining sub-Neptunes.

After Fulton et al. (2017).

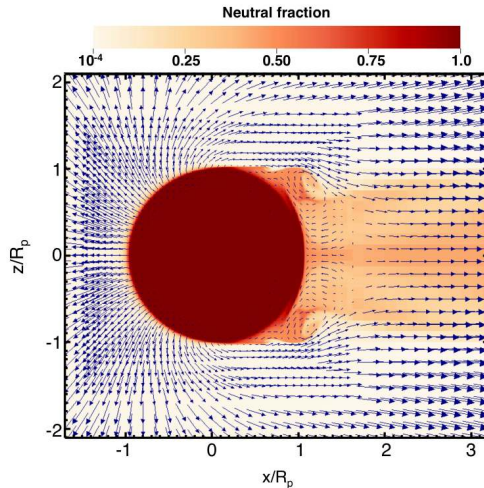
## Energy-limited escape rate

Stellar EUV energy absorbed in the upper atmosphere drives hydrodynamic mass loss in the energy-limited regime.

$$\dot{M}_{\text{EL}} = \eta \frac{\pi F_{\text{EUV}} R_p^3}{G M_p}$$

- Efficiency  $\eta \approx 0.1\text{--}0.2$  converts incident EUV flux into expansion work
- Scales as  $R_p^3$ : extended, low-density atmospheres lose mass most rapidly
- Scales inversely with  $M_p$ : lower planetary gravity enables stronger bulk outflow

Source: Watson et al. (1981); Owen (2019)



Hydrodynamic outflow from a hot Jupiter driven by stellar EUV heating: heated gas accelerates outward into a night-side wake with Kelvin-Helmholtz shear instabilities.

Reproduced from Owen (2019), Fig. 2; simulation from Tripathi et al. (2015).

## Non-thermal escape mechanisms

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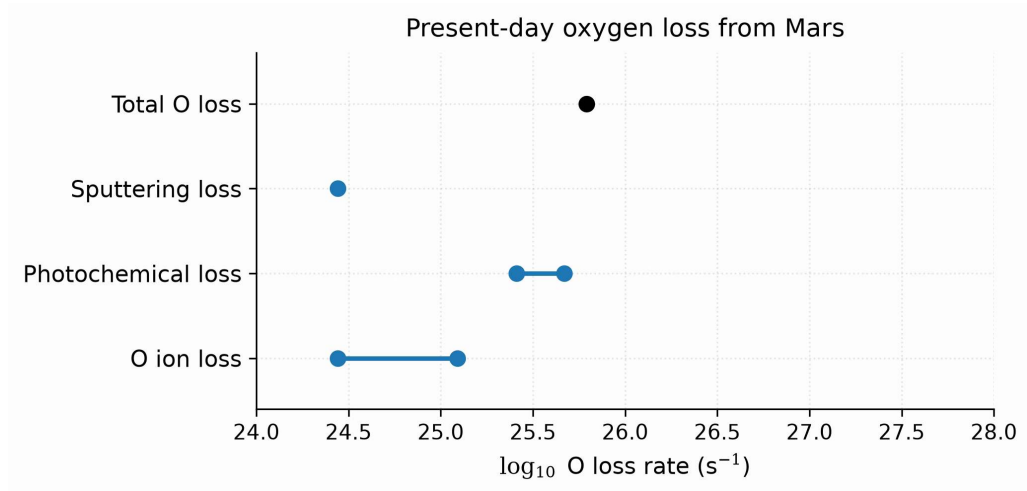
**Sputtering:** Solar wind ions strike upper atmosphere and eject neutral molecules

**Photochemical:** Photodissociation yields suprathermal atoms exceeding  $v_{\text{esc}}$

**Ion pickup:** Photoions are swept away by interplanetary magnetic fields

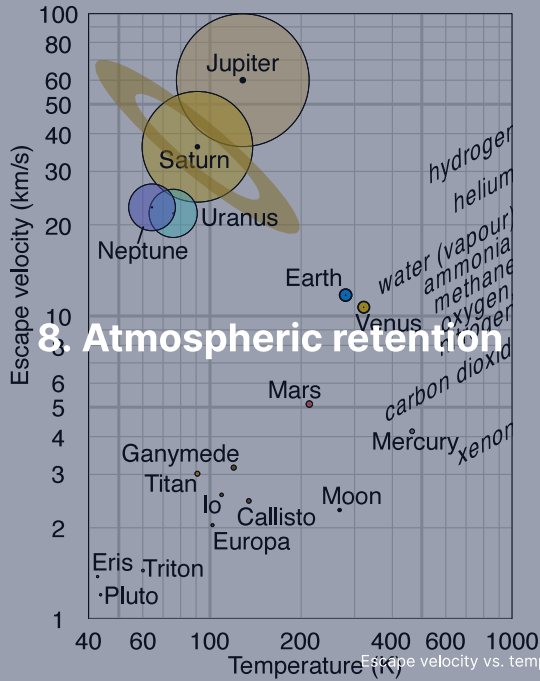
**Impact erosion:** Giant impacts shock and eject atmospheric columns into space

MAVEN at Mars: photochemical loss of oxygen dominates modern escape over sputtering and ion pickup.

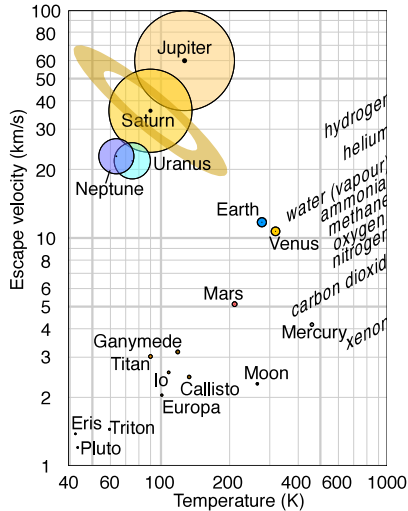


Present-day oxygen loss from Mars by escape channel: photochemical escape dominates modern oxygen loss; summed with hydrogen escape, the total reaches the  $2\text{--}3 \text{ kg s}^{-1}$  MAVEN measures. After Jakosky et al. (2018).





# Atmospheric retention: $v_{\text{esc}}$ vs. $T$ diagram



Credit: Wikimedia, CC BY-SA 4.0

Rule of thumb for long-term retention:

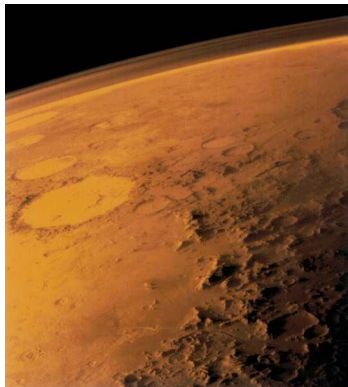
$$v_{\text{esc}} \gtrsim 6 v_{\text{th}}$$

- Gas giants: retain all species
- Earth/Venus: retain heavy species, lose H
- Mars: marginal; non-thermal losses dominate
- Titan: cold  $\rightarrow$  retains  $\text{N}_2$  despite low  $v_{\text{esc}}$
- Moon/Mercury: retain no significant atmosphere

## Mars: a lost atmosphere

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Loss of Mars' magnetic dynamo allowed solar wind stripping to erode an early thick atmosphere down to 6 mbar.

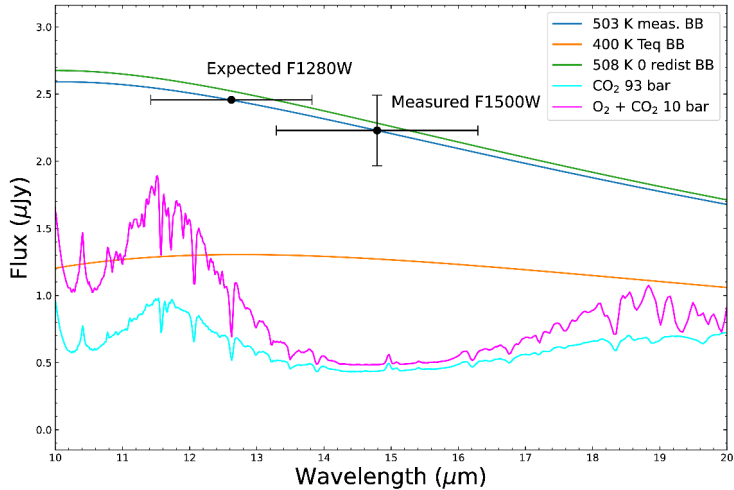


Credit: NASA/JPL (Viking 1, 1976)

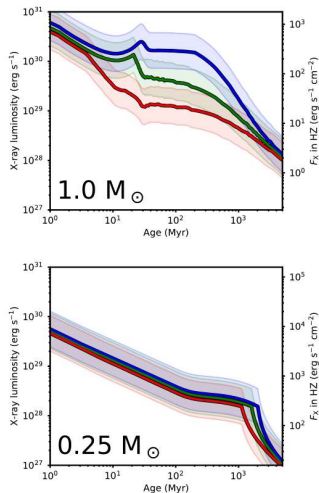
- Today 6 mbar: no stable liquid water, little greenhouse warming
- Valley networks and lakes record a dense early atmosphere
- The dynamo died near ~4.1 Ga and left the atmosphere to the solar wind



## **9. Recent advances & outlook**



JWST/MIRI 15  $\mu\text{m}$  thermal emission matches a 503 K bare rock, ruling out a thick atmosphere on TRAPPIST-1 b. Greene et al. (2023).



Stellar X-ray luminosity evolution for Sun-like stars and M dwarfs: lower-mass stars remain in the saturated high-activity regime for over 1 Gyr, driving prolonged atmospheric stripping on close-in rocky exoplanets. Reproduced from Johnstone et al. (2021), Fig. 11.

# Summary

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1. **Atmosphere types:** primary (captured), secondary (outgassed), tertiary (biogenic)
2. **Hydrostatic balance:**  $\frac{dP}{dz} = -\rho g$  yields exponential pressure decay  $P_0 e^{-z/H}$
3. **Scale height:**  $H = k_B T / (\mu m_u g)$  balances thermal energy against gravity
4. **Vertical structure:** convection sets the lapse rate ( $\Gamma_d = g/c_p$  dry, less when condensing) below the 0.1 bar tropopause
5. **Beer-Lambert law:**  $I = I_0 e^{-\tau}$ ; thermal radiation escapes from photosphere at  $\tau \approx 1$
6. **Greenhouse warming:** optical depth raises surface temperature  $T_s$  above  $T_{\text{eff}}$
7. **Atmospheric escape:** thermal, hydrodynamic, and non-thermal; retention requires  $v_{\text{esc}} \gtrsim 6v_{\text{th}}$

**Next: Lecture 6**

## Questions?

Next: Lecture 6, Atmospheres II: clouds, weather & climate (Mara Attia)

Thu 24 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

<https://ips.formingworlds.space>